

Development of Handover Decision Algorithms in Hybrid Li-Fi and Wi-Fi Networks

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Abstract—A handover decision algorithm in a hybrid Light Fidelity (Li-Fi) and Wireless Fidelity (Wi-Fi) network is investigated in this paper. Li-Fi, a wireless network uses visible light spectrum to provide high-speed indoor data transmission along with illumination. As there is no interference between optical and Radio Frequency (RF) spectrum operating devices, a hybrid Li-Fi/Wi-Fi network (HLWNet) can be explored in order to improve the user Quality of Service (QoS). However, in a HLWNet system setup, user mobility may often prompt frequent handover which as a result, degrades the system throughput. In this paper, we proposed a Fuzzy Logic (FL) and fuzzy rule-based Artificial Neural Network (ANN) handover decision algorithms. The FL based handover algorithm uses input parameters namely the instantaneous Signal to Interference Noise Ratio (SINR), Received Signal Strength (RSS), average SINR and user velocity to decide whether handover needs to be prompted. However, because of the increase in the number of the input parameters which, in turn, increases the number of fuzzy rules, the computational complexity greatly affects the FL system. Thus, it is envisioned that using the learning power of ANN, limited fuzzy rules are generated, which it can be made to learn from the limited rules and be able generalize to make handover decision. Based on the accuracy test conducted, the FL based handover decision algorithm is 1.66 times more accurate than the ANN fuzzy rule-based handover decision algorithm in terms of successfully assigning access points (AP) to users.

Keywords—Artificial neural network (ANN), fuzzy logic (FL), light fidelity (Li-Fi), wireless fidelity (Wi-Fi).

I. INTRODUCTION

The continuous demand for multimedia higher data rates has become evident that the current wireless communication spectrum cannot accommodate the extensively growing mobile data connected devices due to its limited available spectrum [1]. Hence, to be able to meet up with this demand in the medium-to-long term, the need to utilize other available frequency bands is of great important for future mobile communication systems [2].

Light Fidelity (Li-Fi), a subclass of Optical Wireless Communication (OWC) technology has recently been proposed as a potential technology to address the spectrum shortage problem in the Radio Frequency (RF) wireless systems [3]– [10]. Li-

Fi on the other hand can provide high-speed wireless data transmission along with illumination using visible light as its propagation medium [11], [12]. Unlike the Wireless Fidelity (Wi-Fi) technology, Li-Fi uses Light Emitting Diodes (LEDs) fitted with special chips for high-speed data transmission suitable for numerous indoor applications [13].

Therefore, the benefits of these technologies are united in a hybridized/heterogeneous network in order to enhance the bandwidth of the traditional RF systems [14]– [16]. This is because devices in the optical and RF spectrum can operate together without causing interference among users [17]. Thus, by deploring a hybrid Li-Fi/Wi-Fi networks (HLWNet), users Quality of Services (QoS) can be enhanced irrespective of their location within a coverage area [18]– [20]. I.e. the RF and Li-Fi network system benefits from reduced spectrum usage and coverage at dead spots, respectively [21]. However, constant handover is often experienced in a HLWNet system due to the user mobility and the relatively small coverage area of a single Li-Fi access point (AP), and consequently degrades the system throughput performance [22].

In this work, two handover decision algorithms namely a Fuzzy Logic (FL) based and a fuzzy rule-based Artificial Neural Network (ANN) are proposed to address the aforementioned bottleneck problem. Specifically, the problem of handover in a heterogeneous network is addressed using a hybrid Artificial Intelligence (AI) approach. FL has proven to be one of the powerful AI techniques that assists in decision making in various fields [23]– [25]. However, because the number of rules generated in a FL system which depends on the fuzzy membership function (MF) and the input parameters, these rules increases as these two parameters increases. Thus, it is envisioned that using the learning power of ANN, limited fuzzy rules are generated and that ANN can be made to learn these limited rules and thereby making an intelligence handover decision in heterogeneous networks (HetNets). Furthermore, performance analysis of the two proposed handover decision algorithms is presented in this paper.

The remaining part of this paper is organized as follows: Section II provides overview of related works. Section III

describes the system model design for the proposed FL-based handover decision algorithm. Fuzzy rule-based ANN handover decision algorithm is presented in Section IV. Data used for training and validation of the proposed schemes are summarized in Section V. Simulation results and conclusions are discussed in Section VI and VII, respectively.

II. RELATED WORK

A fast and accurate ANN-based handover decision algorithm in HetNets consisting of three layers, namely input, hidden and output layer, was proposed by Partha in [26]. The author used an iterative neural back-spreading system using parameters such as obtained signal strength (RSS) and traffic density in both current and target networks. The process begins with the neural network's last layer and works backward through the layers before the first layer is reached. As a result, the output and the errors are computed and the weights on the output unit are altered. Results obtained by the author indicate that the decisions on handovers are taken appropriately and that the number of fluctuations as well as the total number of handovers are minimal. However one of the drawbacks of this system is the delay in the learning procedure.

Horrich et al. in [27] combined the power of ANN and a fuzzy logic controller (FLC) to propose an efficient handover decision algorithm. The authors used RSS, user velocity and network load as handover decision parameters. In the model, the neural network gets these parameters from the FLC and then a preliminary selection of networks is done before initiating the vertical handover (VHO). The target networks with signal level and network load above a predetermined threshold are filtered in the preliminary network and the best is selected. The proposed scheme reduces the FLC complexity and the processing duration. However, when the network conditions are altered, the proposed scheme may not easily adapt to these changes. If the network load increases, for example, the overall performance of the proposed system deteriorates. Therefore, the neural network has to be fast in order to tackle the problem.

Attaullah et al. in [28] proposed an intelligent FL-based vertical handover algorithm to improve the QoS in a HetNet system. In order to avoid interruption of connection, the authors employed make before break handover strategy. The model uses Rss, available bandwidth and network load as the parameters of the handover decision. An entity event collector that stores the states of every interface variable that can be used to further process the final output from the fuzzy expert system is used. This system takes the inputs parameters as crisp inputs from the event collector and tests them according to pre-defined to handover rules. Using these rules, the crisp input is then evaluated. The aggregate output is defuzzified and a crisp output that was used for the handover is obtained. However, detailed justifications for the handover decision process were not given by the authors.

Tu et al. in [29] presented a fuzzy inference system capable of selecting an appropriate network. The input parameters used for the handover decision were RSS, available bandwidth and distance from the access point. The input parameters are fed

into the fuzzifier, which converts them to fuzzy sets using a membership function. The fuzzy sets are first added to the fuzzy inference engine so that the IF-THEN rules can be applied in order to acquire the fuzzy decision sets. The authors performed handover between WLAN and WIMAX to analyze the performance of the proposed scheme and as a result, succeeded in reducing the delay in handover and the loss of packets.

Ioannis et al. in [30] proposed a modified TOPSIS fuzzy technique to determine the best balance between network resources and energy consumption in the network selection process in HLWNets. In order to select an optimal network, the authors used a utility function that takes the network condition such as the QoS, energy and user preferences as parameters. The proposed model is able to address the inability problem of ranking the networks where there are many networks. The authors, however, were silent on details about when a handover would be triggered and how the scanning of the network was performed.

Moreover, a network selection scheme based on service characteristics and user preferences for HetNets was proposed by Gen et al. in [31]. To measure the utility value associated with each network for different services, the authors utilized the power of the utility functions. The objective and subjective weights of the network attributes were calculated using the entropy method and the fuzzy-analytic hierarchy process (FAHP), respectively. Finally, the authors used TOPSIS, Multiplicative Exponent Weighting (MEW) and Simple Additive Weighting (SAW) to calculate the score of each candidate network, and used the scores to obtain the ranking of candidate networks. Results obtained by the authors show a reduction in the number of network handovers and allow users to choose the most suitable network based on the different service characteristics. However, authors didn't consider minimizing network latency performance in this model.

Authors in [32] presented a FL-based dynamic handover scheme to mitigate the effects of handover in a HLWNet. The authors were able to decrease the number of handover by appropriately allocating users to either a Li-Fi or RF AP using the proposed scheme. The FL system model consists of four steps: fuzzification, rule evaluation, defuzzification and decision-making. In order to determine whether a handover is required or not, the channel state information, user speed and data rate are used as the inputs in the FL system. Results from the simulation show that in terms of less data rate degradation, the proposed FL-based algorithm outperforms the conventional load balancing algorithm. Moreover, a 40% performance improvement in terms of QoS and data rate of a user is achieved using the proposed model. However, this scheme suffers from computational complexity as the fuzzy inputs increases which affect the overall performance of the system.

Moreover, authors in [33] proposed a fuzzy decision-making strategy for VHO in HetNets. As the decision criteria, the authors used Rss, Variation of the Rss (VRss), bandwidth, traffic status and handover preference in the model. The authors

used Rss, Variation of the Rss (VRss), bandwidth, traffic status and handover preference as the decision parameters. A five-step handover decision scheme namely handover initiation, context preparation, fuzzy conversion, preliminary selection and Multi Attribute Decision Making (MADM) were proposed by the authors. The handover initiation feeds the parameters of the network into a fuzzifier. The fuzzifier then converts the input into fuzzy sets and assigns values such as *Weak*, *Medium* and *Strong*. At the preliminary handover selection stage, unsuitable candidates networks are filtered out and handover is done to the best network after the defuzzification stage. The proposed decision strategy used in this model adequately deals with handover triggering by taking into account forced voluntary and forced handovers.

A vertical handover (VHO) scheme for heterogeneous system was also investigated by Fang et al. in [34]. By formulating the VHO as a Markov decision process problem and adopting a dynamic approach, the authors were able to obtain a trade-off solution that decreases the switching cost of the system. In order to decide if a VHO should be prompt or no, this scheme takes into account the conditions of the network channels and the current length of the packet queue. The proposed model is able to save up to 50% of the signaling cost without a significant loss in system performance.

Many other approaches to find an efficient handover decision making algorithm in HetNets have also been investigated by authors in [35]- [41].

III. SYSTEM DESIGN MODEL

In order to implement a FL technique, three stages namely: 1) fuzzification, 2) fuzzy inference and 3) defuzzification are required [42]. These three stages are further discussed in the next subsections. Fig. 1 illustrates the simple block diagram of a handover decision system based on FL.

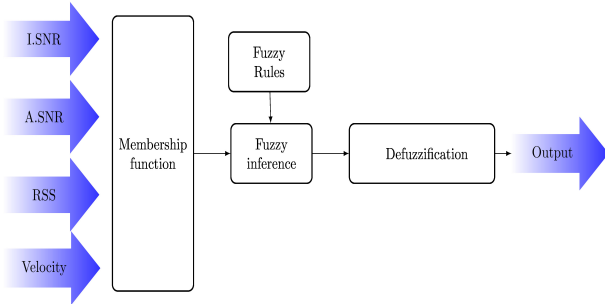


Fig. 1. Block diagram of the fuzzy logic system.

A. Fuzzification Stage for FL based Handover Algorithm

This stage converts the input data of the FL system to crisp values using the MF as depicted in Fig. 2. In this work, the instantaneous SINR (I.SINR), average SINR (A.SINR), RSS and the user velocity are used as the inputs of the FL system. Users upload their input information without any delay. Each of the input is denoted by three grades namely Low, Medium, and High [32]. A value between 0 and 1

is assigned to represent how well the input fits each grade. Moreover, a triangular membership function adopted in this work is denoted by:

$$f_{\text{TRI}}(x; a, b) = \begin{cases} 0, & x \leq a \\ \frac{x-a}{b-a}, & a < x \leq b \\ 1, & x > b \end{cases} \quad (1)$$

and the fuzzified values of the three grades is expressed as:

$$\text{Low} : 1 - f_{\text{TRI}}(x; c_1, c_2) \quad (2)$$

$$\text{Medium} = \begin{cases} f_{\text{TRI}}(x; c_1, c_2), & x \leq c_2 \\ 1 - f_{\text{TRI}}(x; c_1, c_2), & x > c_2 \end{cases} \quad (3)$$

$$\text{High} : 1 - f_{\text{TRI}}(x; c_2, c_3) \quad (4)$$

where x is the input and c_1 , c_2 and c_3 are the break points of the MF for each input. Additionally, each user has a 12 crisp fuzzy set values to describe all the four inputs.

B. Fuzzy Inference Stage for FL based Handover Algorithm

In this stage, the fuzzy rules are combined with the MF in order to obtain the fuzzy output. The AP allocation output for each user has two possibilities: Li-Fi or Wi-Fi AP. The rules adopted in this work are presented in Table 1. These rules act as the AP allocation guidelines that allocate a user to either Li-Fi or Wi-Fi AP. For example, in rule no. 1, the user was assigned to Li-Fi AP because the I.SINR of the user is high and the corresponding velocity of the user is low. Again, we see that in rule no. 10, the user was assigned to Li-Fi AP even though the I.SINR of the user is low. This may be as a result of line of sight blockage and because the A.SINR is high, the user is still allocated to Li-Fi AP. However, we see that in rule no. 14 and 15, the user was allocated to both Wi-Fi APs because the velocity is high and medium, respectively, while the A.SINR and I.SINR are both low in this case thereby making it logical to assign the user to Wi-Fi AP.

TABLE 1. Fuzzy rules

Rule No.	I.SINR	A.SINR	Rss	Velocity	AP Allocation
1	High	-	-	Low	Li-Fi
2	-	High	-	Low	Li-Fi
3	-	High	High	Low	Li-Fi
4	High	Medium	Low	Low	Li-Fi
5	High	High	Low	Low	Li-Fi
6	Medium	High	Low	Low	Li-Fi
7	High	High	Low	Medium	Li-Fi
8	High	High	-	Low	Li-Fi
9	High	Medium	Low	Low	Li-Fi
10	Low	High	Low	Low	Li-Fi
11	-	-	High	High	Wi-Fi
12	Medium	Low	High	Medium	Wi-Fi
13	-	Low	Medium	High	Wi-Fi
14	Low	Low	High	High	Wi-Fi
15	Low	Low	Medium	Medium	Wi-Fi
16	-	-	-	High	Wi-Fi
17	Low	-	High	High	Wi-Fi
18	Low	Low	-	-	Wi-Fi
19	Low	Low	-	High	Wi-Fi
20	-	Low	-	Medium	Wi-Fi

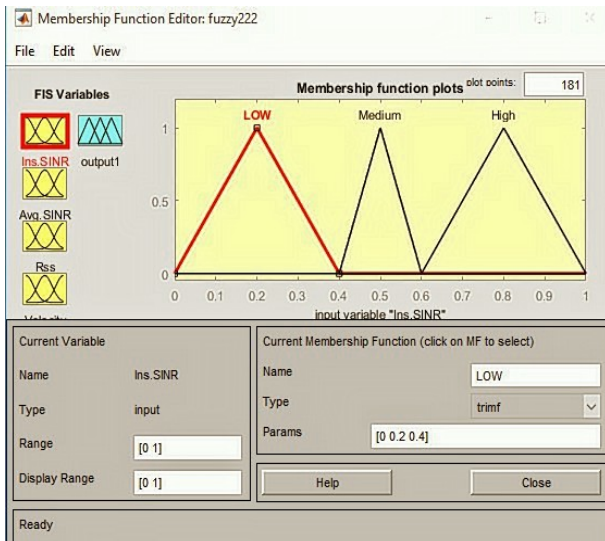


Fig. 2. Membership function of the input parameters.

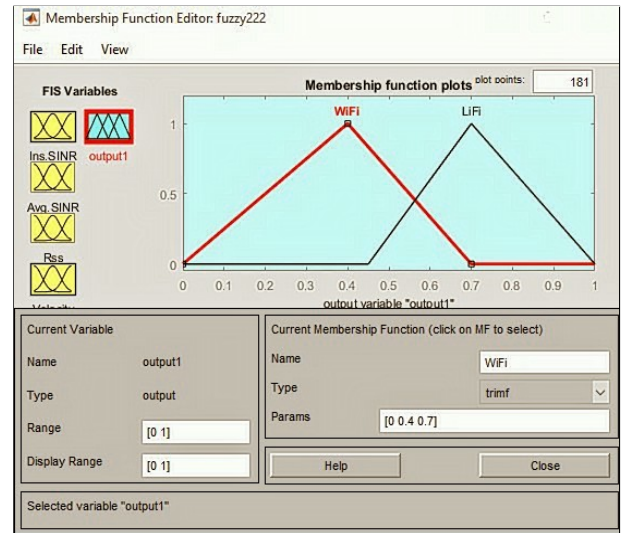


Fig. 3. Membership function of the output parameters.

C. Defuzzification Stage for FL based Handover Algorithm

The centroid defuzzification was used to obtain a numerical value after the output has been obtained as shown in Fig. 3. This stage makes decision on AP allocation i.e. decision on which network to handover to is decided in this stage based on the output obtained. After the output has been obtained from the defuzzification stage, the algorithm depicted in Fig. 3 was used to finally connect users to AP based on their output obtained. The output ranges from 0 to 1. The higher the output, the higher the possibility to handover to Li-Fi AP. The handover decision algorithm is summarized as:

- If the output is less than 0.45 the user will be connected to Wi-Fi AP. If the user is currently connected to Li-Fi THEN a handover to Wi-Fi is done; otherwise no handoff.
- If the output is greater than 0.70 the user will be connected to Li-Fi AP. If the user is currently connected to Wi-Fi THEN a handover to Li-Fi is done; otherwise no handoff.
- If the output is between 0.45 and 0.7, the user continues to maintain the network it's currently connected to. This could reduce unnecessary handover and it solves the ping pong effect associated with hybrid Wi-Fi and Li-Fi networks

The flowchart used in developing the FL-based handover decision algorithm is illustrated in Fig. 4.

IV. DEVELOPMENT OF FUZZY RULE-BASED ANN HANDOVER DECISION ALGORITHM

This section presents the development of fuzzy rule-based ANN handover decision algorithm. The ANN gets its input and output from the fuzzy rule and then makes its decisions based on the inputs. The inputs layer in a typical ANN architecture contains four neurons, the output layer contains 2 neurons corresponding to input and output parameters respectively while the hidden layer contains varying layers as depicted in

Fig. 5. ANN basically learns from experience, example and by analogy. These learning capabilities can also improve the performance of intelligent the system overtime.

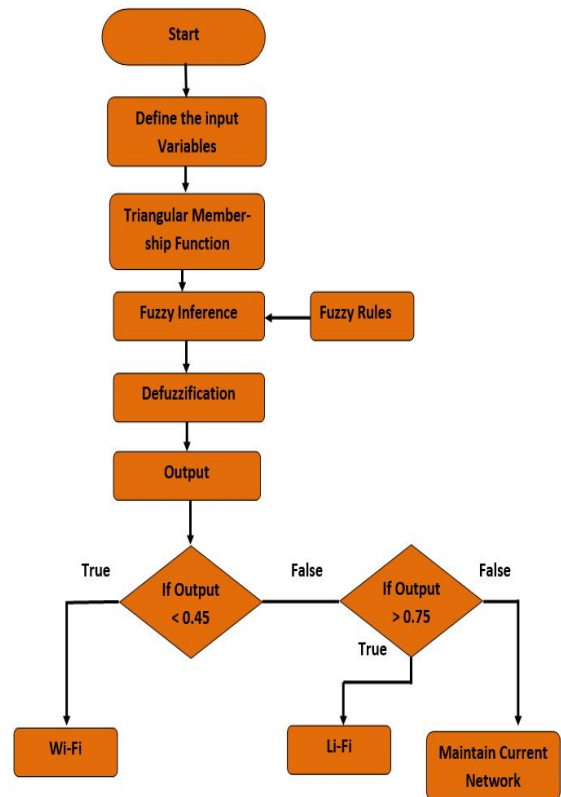


Fig. 4. Flowchart of the developed FL based handover decision algorithm.

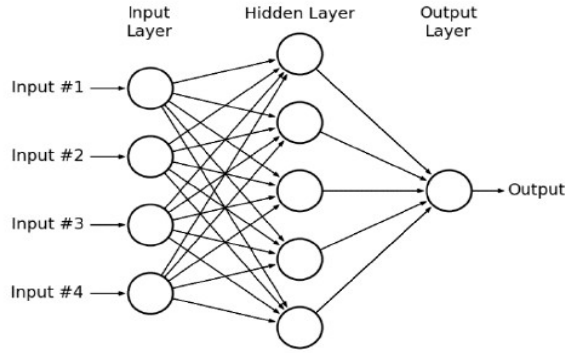


Fig. 5. A typical ANN architecture [43].

A. Defining Input and Output Parameters

The fuzzy rules used in section III will be the input and output parameters of the ANN handover decision algorithm. After getting the data from the fuzzy rule and converting them to binary numbers using Table 2 and 3, the ANN makes handover decisions. As the number of inputs to the FL increases, so the fuzzy rules increases which poses a huge task on the system. Since increase in number of fuzzy rule leads to high complexity, ANN was trained to learn from the fuzzy rules and based on that, a handover decision will be made. This solves the problem of high system complexity.

TABLE 2.

A table used to convert fuzzy rules to ANN input values

High	1	0	0
Medium	0	1	0
Low	0	0	1
xxx	0	0	0

TABLE 3.

A table used to convert fuzzy rules to ANN output values

o/p1	o/p1	Outcome
1	0	Wi-Fi
0	1	Li-Fi

V. DATA USED FOR TRAINING AND VALIDATION

After converting the fuzzy rule binary, Table 4 was then used to train the ANN. Trained and untrained data used in validating the ANN algorithm are given Table 5 and 6, respectively, was used to test the performance of the ANN. Also, the subscript M and L used in Table 4, 5, and 6, stands for the most significant bit and the least significant bit, respectively. Additionally, the accuracy test used in analyzing the performances of the proposed handover algorithms is given by Eqn. (5)

$$\text{Accuracy test} = \frac{\text{No. of successful Handover}}{\text{Total number of Handover}} * 100\% \quad (5)$$

TABLE 4. Input and output values used in training the ANN

Rule No.	I.SINR _L	I.SINR	I.SINR _M	A.SINR _L	A.SINR	A.SINR _M	Rss _L	Rss	Rss _M	Vel _L	Vel	Vel _M	o/p1	o/p1
1	0	0	1	0	0	0	0	0	0	1	0	0	0	1
2	0	0	0	0	0	1	0	0	0	1	0	0	0	1
3	0	0	0	0	0	1	0	0	1	1	0	0	0	1
4	0	0	1	0	1	0	1	0	0	1	0	0	0	1
5	0	0	1	0	0	1	1	0	0	1	0	0	0	1
6	0	1	0	0	0	1	1	0	0	1	0	0	0	1
7	0	0	1	0	0	1	1	0	0	0	1	0	0	1
8	0	0	1	0	0	1	0	0	0	1	0	0	0	1
9	0	0	1	0	1	0	1	0	0	1	0	0	0	1
10	1	0	0	0	0	1	1	0	0	1	0	0	0	1
11	0	0	0	0	0	0	0	0	1	0	0	1	1	0
12	0	1	0	1	0	0	0	0	1	0	1	0	1	0
13	0	0	0	1	0	0	0	1	0	0	0	1	1	0
14	1	0	0	1	0	0	0	0	1	0	0	1	1	0
15	1	0	0	1	0	0	0	1	0	0	1	0	1	0
16	0	0	0	0	0	0	0	0	0	0	0	1	1	0
17	1	0	0	0	0	0	0	0	1	0	0	1	1	0
18	1	0	0	1	0	0	0	0	0	0	0	0	1	0
19	1	0	0	1	0	0	0	0	0	0	0	1	1	0
20	0	1	0	0	0	0	0	0	0	0	0	1	1	0

TABLE 5. Trained data used to validate the ANN

Rule No.	I.SINR _L	I.SINR	I.SINR _M	A.SINR _L	A.SINR	A.SINR _M	Rss _L	Rss	Rss _M	Vel _L	Vel	Vel _M
1	0	0	1	0	0	0	0	0	0	1	0	0
2	0	0	0	0	0	1	0	0	0	1	0	0
3	0	0	0	0	0	1	0	0	1	1	0	0
4	1	0	0	1	0	0	0	0	0	0	0	1
5	0	0	0	1	0	0	0	0	0	0	1	0

TABLE 6. Untrained data used to validate the ANN

Rule No.	I.SINR _L	I.SINR	I.SINR _M	A.SINR _L	A.SINR	A.SINR _M	Rss _L	Rss	Rss _M	Vel _L	Vel	Vel _M
1	0	0	1	0	1	0	0	0	1	0	0	1
2	0	0	1	1	0	0	0	0	1	0	0	1
3	1	0	0	1	0	0	1	0	0	1	0	0
4	0	0	1	0	0	1	0	0	1	0	0	1
5	0	0	1	0	0	0	0	0	0	1	0	0

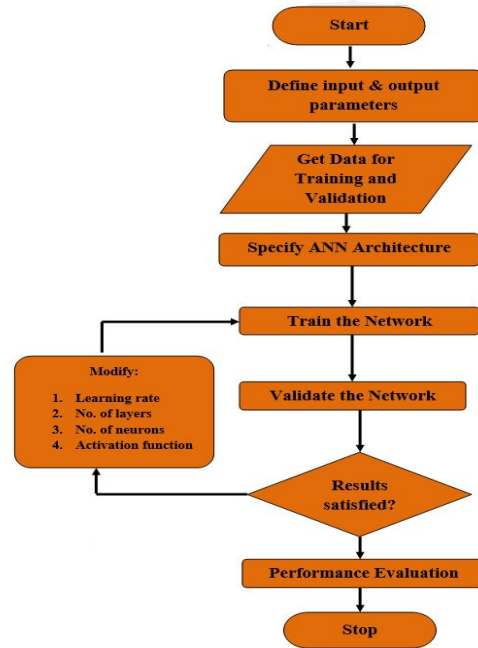


Fig. 6. Flowchart of the Designed and developed fuzzy rule-based ANN handover decision algorithm.

VI. RESULTS AND DISCUSSIONS

A. Performance Analysis of FL Based Handover Decision Algorithm

After uploading the input parameters to the FL, an output is obtained using the fuzzy handover decision inference engine (see Fig. 7). Based on the obtained output, a decision is made to connected the users to either Li-Fi or Wi-Fi AP.

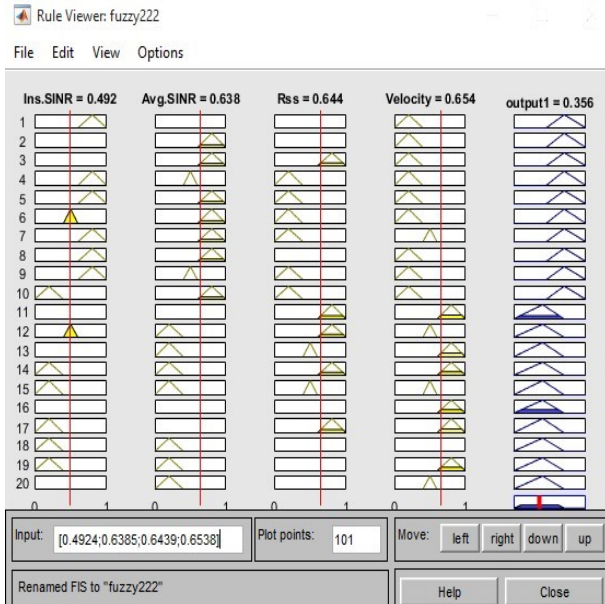


Fig. 7. Fuzzy handover decision inference engine.

Table 7 presents the performance analysis of the developed FL rule-based decision algorithm. For every input, a handover decision value was obtained based on that, a handover is done to either Li-Fi or Wi-Fi as the case may be. For I.SINR of 0.644, A.SINR of 0.177, Rss of 0.750 and velocity of 0.823, a handover decision value of 0.32 was obtained and the user was connected to a Wi-Fi network based on the algorithm. In S/No. 3 for example, the input values obtained for the I.SINR, A.SINR, Rss and velocity are 0.5, 0.3, 0.2, 0.6, respectively, and a handover decision value of 0.572 was obtained which corresponds to ‘users should maintain their network’ according to the algorithm in Fig. 4. I.e no handover should be done if the handover decision value falls between 0.45 to 0.70. This helps to prevent unnecessary handover which degrades the performance of the network. Hence this technique also tackles the ping pong effect associated with hybrid networks of Li-Fi/Wi-Fi. And lastly, in the last experiment of S/No. 5, a handover value of 0.727 was obtained and accordingly the user was connected to Li-Fi network.

B. Performance Analysis of Fuzzy Rule-Based ANN Handover Decision Algorithm

The performance analysis of the fuzzy rule-based ANN handover decision algorithm was done using the trained and

untrained data given by Table 5 and 6, respectively and the results were tabulated in Table 8. A comparison was done between the desired output and the actual output of the ANN and their errors were also computed to verify whether the ANN was able to learn or not. The results clearly show that ANN was able to learn and take appropriate decision both in the case of trained and untrained data with minimal error. Hence, a fuzzy rule-based ANN handover decision algorithm was developed which has the capability of learning from the fuzzy rules and to make handover decisions with negligible errors. Moreover, Fig. 8 shows the plot performance of the ANN trained network and the performance of the trained fuzzy rule-based ANN handover decision algorithm with the mean squared error on the y-axis and the number of Epochs on the x-axis. The graph obtained is a decaying graph which implies that ANN was able to learn and can make handover decisions with minimal error.

For an iteration p as shown in Table 8, $Y_d(p)$ and $Y(p)$ represents a desired and actual output, respectively. Correspondingly, the error is denoted as:

$$e(p) = Y_d(p) - Y(p) \quad (6)$$

TABLE 7. Performance analysis of the designed and developed fuzzy logic based handover decision algorithm

S/No	I.SINR	A.SINR	Rss	Velocity	Handover Decision Value	Decision
1	0.644	0.177	0.750	0.8230	0.342	Handover to Wi-Fi if current network is Li-Fi else no Handover
2	0.886	0.885	0.285	0.7650	0.743	Handover to Li-Fi if current network is Wi-Fi else no Handover
3	0.500	0.300	0.200	0.600	0.572	MN maintains its network. There is no need for handover. This prevents ping pong effect
4	0.356	0.315	0.644	0.792	0.344	The user will be connected to Wi-Fi network if the user's current network is Li-Fi else no handover is done
5	0.705	0.869	0.402	0.146	0.727	User is connected to Li-Fi network

C. Performance Evaluation

This section presents the performance comparison between the two proposed handover decision algorithms and the results are tabulated in Table 9 and 10. The ANN was tested with untrained data to see the decisions it will take as against the FL and a comparison was made. Table 9 shows the simulation results of the two proposed algorithms i.e, a comparison was made between the handover decision of the FL algorithm and ANN algorithm. In rule No. 1 and rule No. 4 (See Table 9),

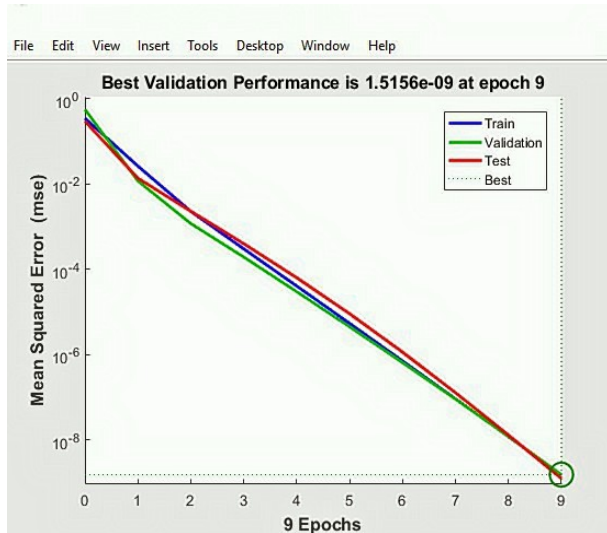


Fig. 8. Fuzzy rule-based ANN performance showing best validation mean squared error.

the two algorithms made different AP allocations while in rule No. 2, 3 and rule No. 5, the AP allocation of the two algorithms was the same. As a result of these variations in terms of AP allocations, a comparison in terms of what the algorithm allocated and the expected allocated was done and tabulated in Table 10 with a successful handover denoted by 1 and a handover failure denoted by 0.

VII. CONCLUSION

Two handover decision algorithms based on Fuzzy Logic (FL) and fuzzy rule Artificial Neural Network (ANN) were designed and developed to tackle the frequent handover experienced in the hybrid Li-Fi /Wi-Fi network (HLWNet). The former uses the instantaneous SINR (I.SINR), average SINR (A.SINR), received signal strength (RSS) and the velocity of the user as the inputs parameters of the FL system combined with the fuzzy rules to obtain the fuzzy output that is used to allocate a user to either a Li-Fi or Wi-Fi access point (AP). The latter uses the leaning power of ANN based on the input and the obtained output from the fuzzy rule and then makes its decisions of allocating a user to either Li-Fi or Wi-Fi AP. The performance analysis of the two proposed algorithms was carried out. It was observed that the FL based handover decision algorithm was capable of making appropriate decisions based on the rules while the performance of the fuzzy rule-based ANN handover decision algorithm was simulated using trained data from the fuzzy rules. It was also observed that the fuzzy rule-based ANN handover decision algorithm was capable of learning from the fuzzy rules and make handover decisions with minimal errors. Lastly, an accuracy test was done on the two algorithms with FL based handover algorithm having 100% handover success rate while the fuzzy rule-based ANN handover decision algorithm having 60%.

TABLE 8. Results of the fuzzy rule-based ANN handover decision algorithm using trained data

Inputs										Desired Output(Yd)	Actual Output(Y)	Error		Conclusion
0	0	1	0	0	0	0	0	0	1	0	1	0.000	1.000	Li-Fi
0	0	0	0	0	1	0	0	0	1	0	1	0.000	0.999	Li-Fi
0	0	0	0	0	1	0	0	1	1	0	1	0.000	0.999	Li-Fi
1	0	0	1	0	0	0	0	0	1	1	0	1.000	0.000	Wi-Fi
0	0	0	1	0	0	0	0	0	1	0	1	1.000	0.000	Wi-Fi

TABLE 9. Simulation results of the two proposed algorithms

Rule No.	Inputs to fuzzy logic system				Output	Inputs to FL rule based ANN										Output		
1	High	Medium	High	High	Wi-Fi	0	0	1	0	1	0	0	0	1	0	0	1	Li-Fi
2	High	Low	High	High	Wi-Fi	0	0	1	1	0	0	0	0	1	0	0	1	Wi-Fi
3	Low	Low	Low	Low	Wi-Fi	1	0	0	1	0	0	1	0	0	1	0	0	Wi-Fi
4	High	High	High	High	Wi-Fi	0	0	1	0	0	1	0	0	1	0	0	1	Li-Fi
5	High	-	-	Low	Li-Fi	0	0	1	0	0	0	0	0	0	1	0	0	Li-Fi

TABLE 10. Output comparison between the two proposed algorithms

Rule No.	FL Output	ANN Output	Expected Output	Handover Success FL	Handover Success ANN
1	Wi-Fi	Li-Fi	Wi-Fi	1	0
2	Li-Fi	Li-Fi	Li-Fi	1	1
3	Wi-Fi	Wi-Fi	Wi-Fi	1	1
4	Wi-Fi	Li-Fi	Wi-Fi	1	0
5	Wi-Fi	Wi-Fi	Wi-Fi	1	1
			Total	5	3

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